Basics of Spatial Regression

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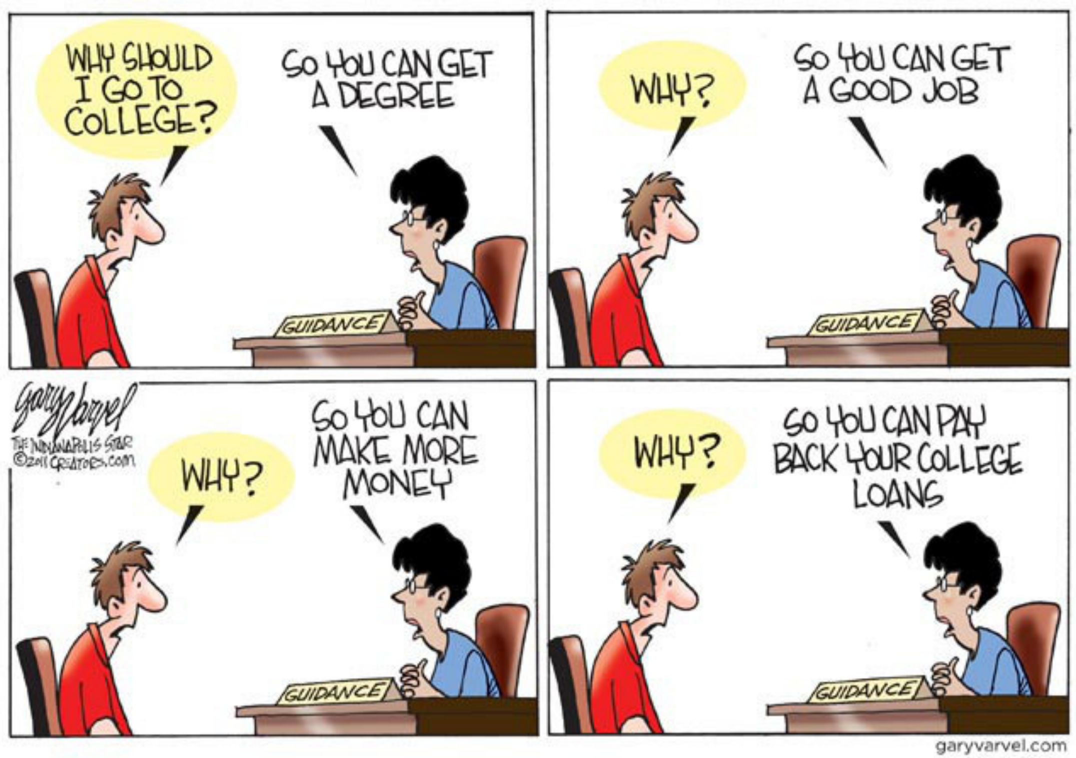
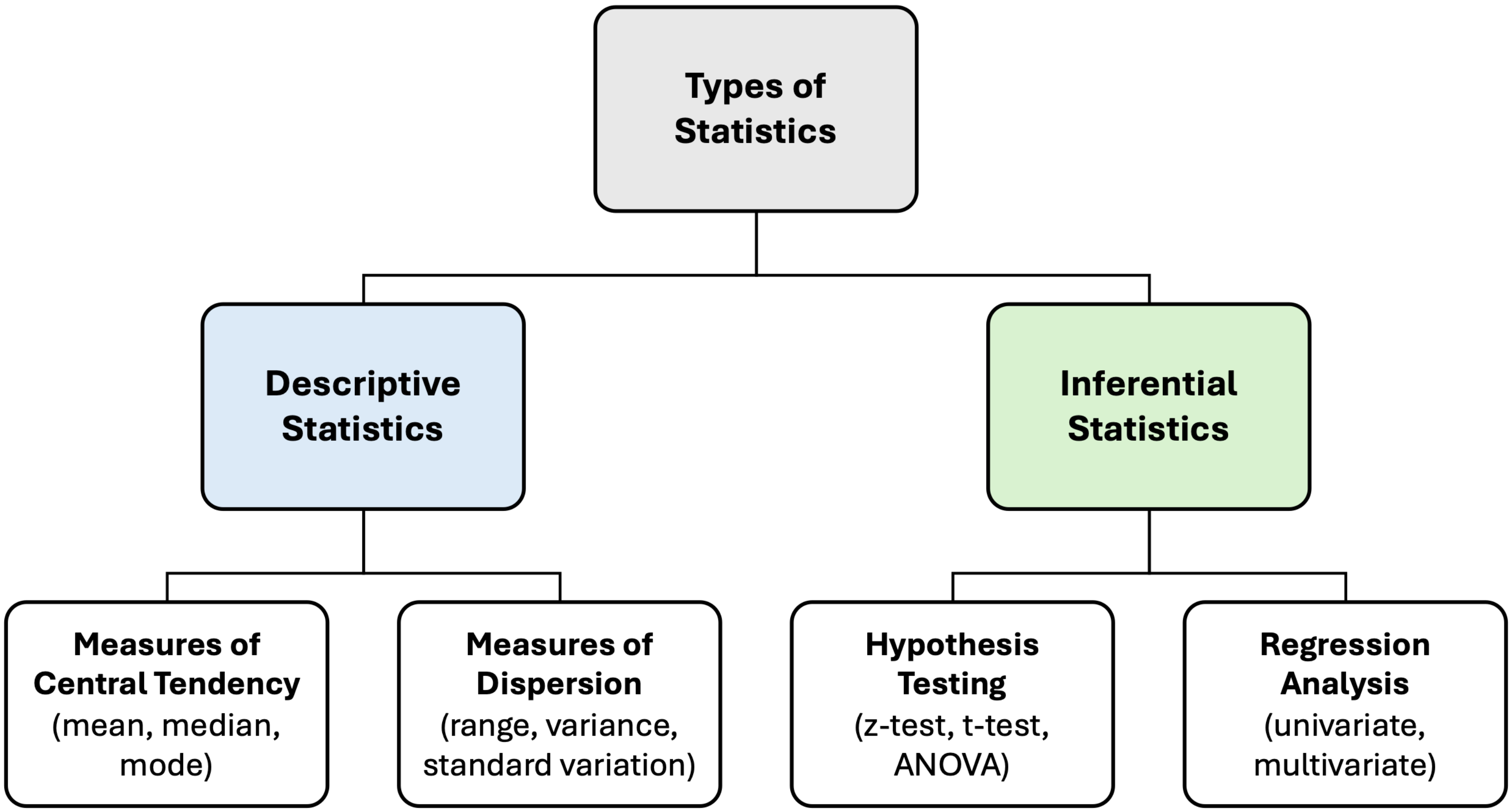
DAEN 489 ISEN 489 Spring 2026

ENGINEERING
TEXAS A&M UNIVERSITY

Question

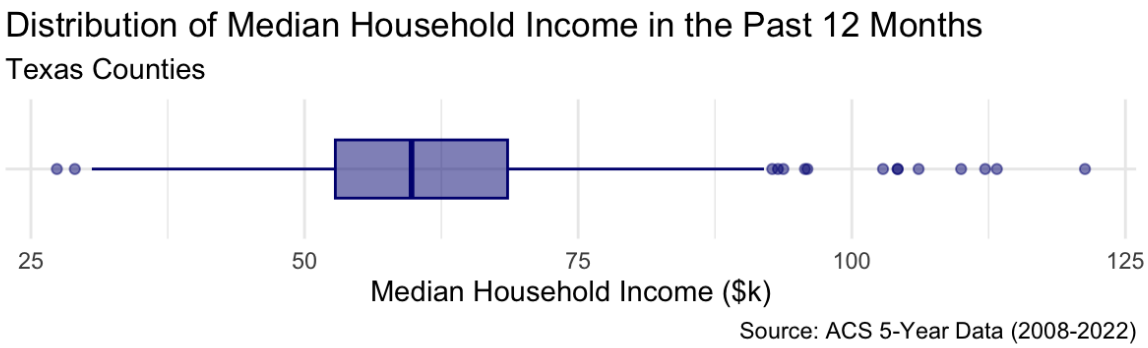
- We all know: "*Go to college, get a better job, make more money*"
- Let us look at all 254 Texas counties
- See if education is a driver of wealth

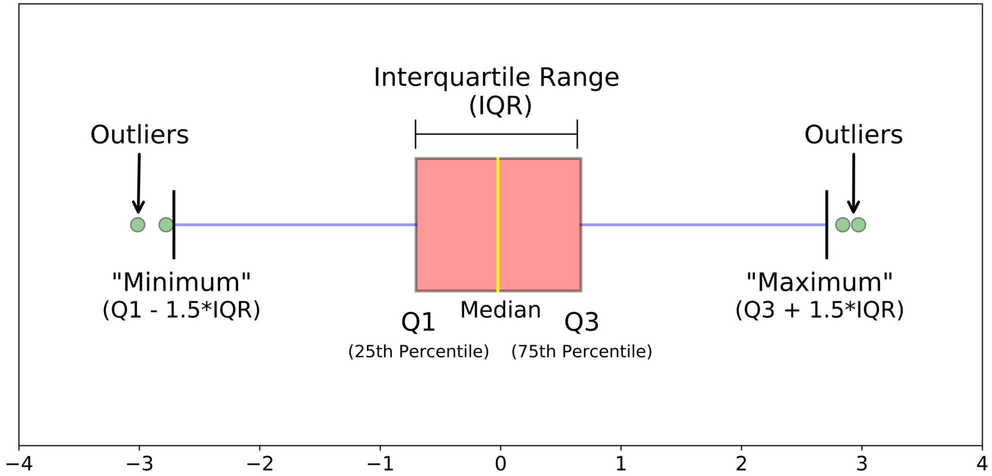



Types of Statistics
Descriptive Statistics
Inferential Statistics
Measures of Central Tendency (mean, median, mode)
Measures of Dispersion (range, variance, standard variation)
Hypothesis Testing (z-test, t-test, ANOVA)
Regression Analysis (univariate, multivariate)

Data

- Pulled 2022 ACS 5-year census data.
- **Dependent Variable**: Median Household Income





Data

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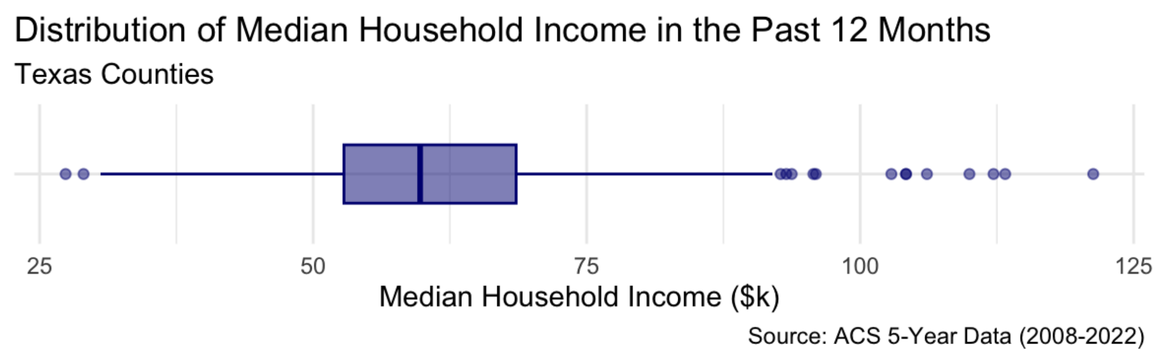
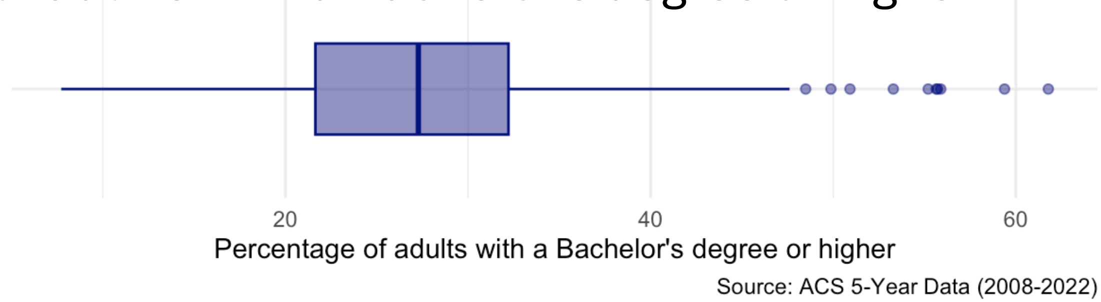
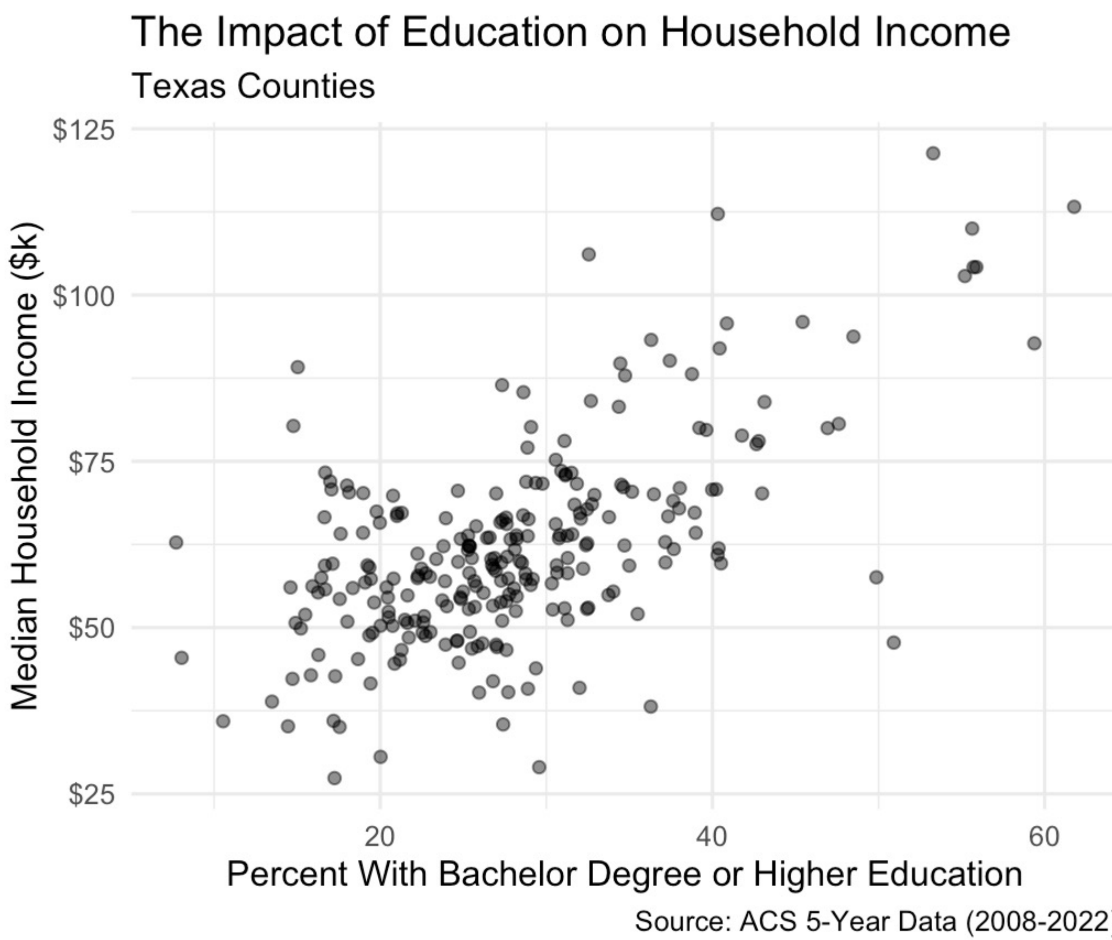
- **Dependent Variable**: Median Household Income



- **Independent Variable**:
  Percentage of adults with a Bachelor's degree or higher




The Impact of Education on Household Income
Texas Counties
Median Household Income ($k)
$125
$100
$75
$50
$25
20
40
60
Percent With Bachelor Degree or Higher Education
Source: ACS 5-Year Data (2008-2022)

Baseline (OLS)

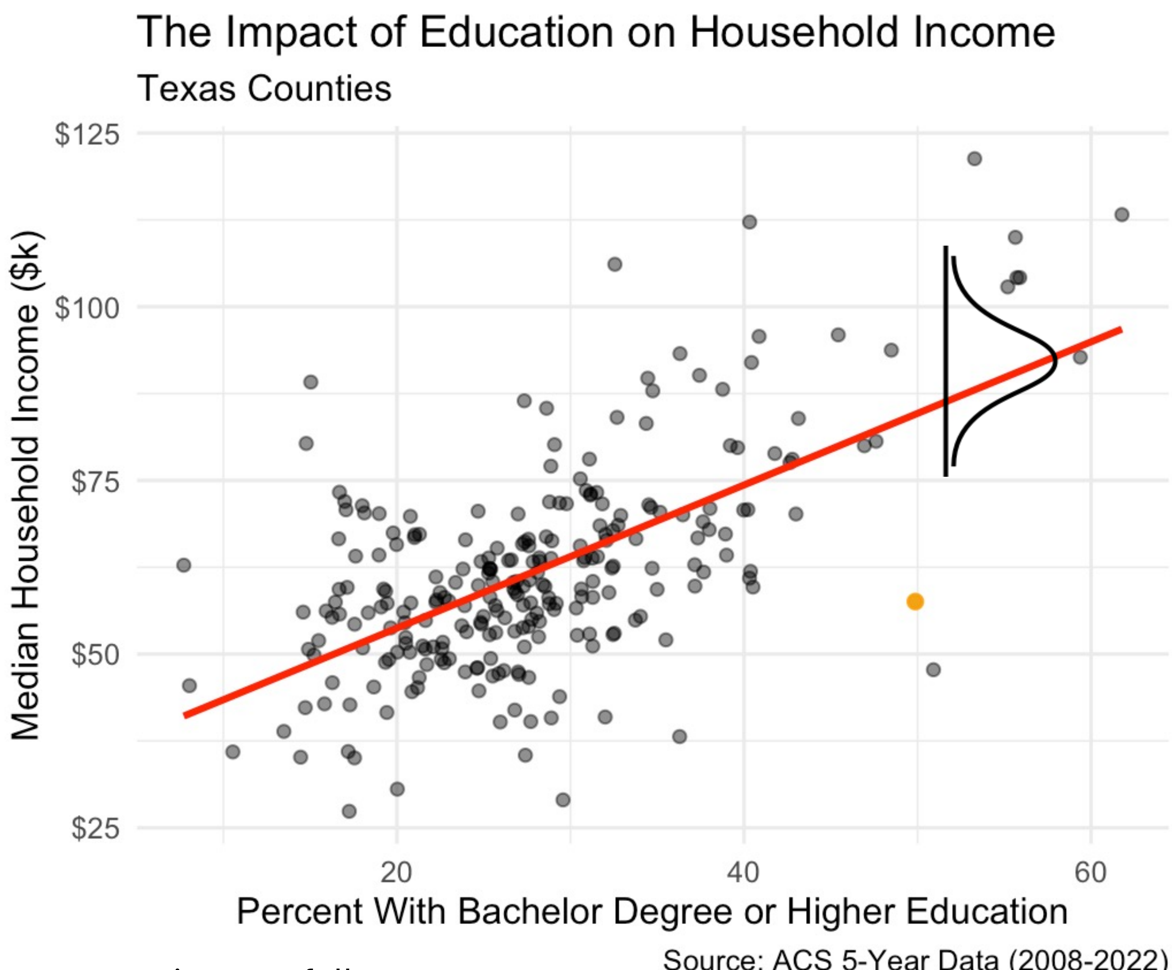
- First model, a simple linear regression:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

  - Y: Median Household Income.
  - X: % with a Bachelor's degree or higher.
- This model assumes ϵ (the error) is independent.

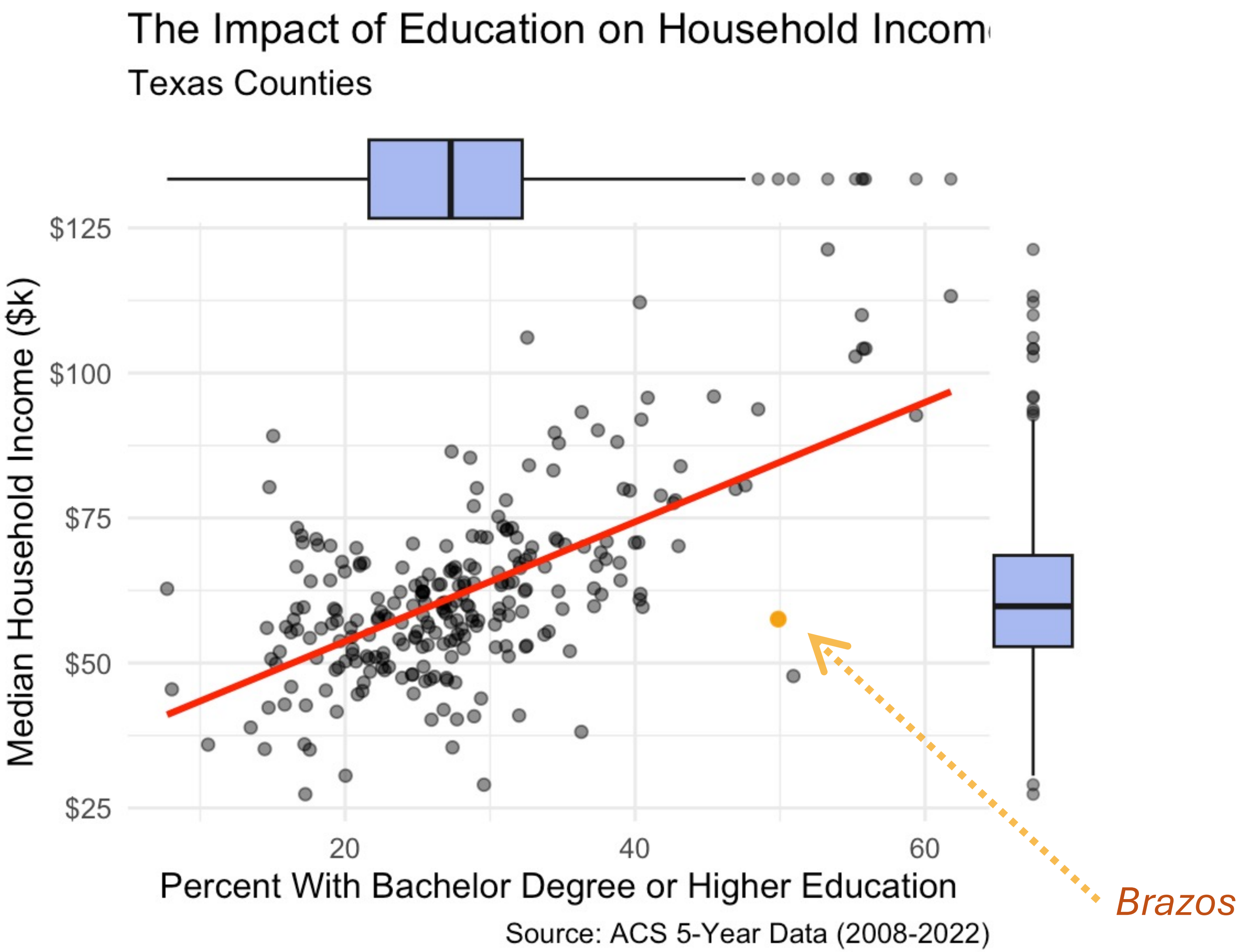
	Dependent variable:	
	med_income_k	
	(1)	(2)
pct_with_diploma		1.03***
Constant	62.04***	33.13***
Observations	253	253
R2	0.00	0.38
Residual Std. Error	15.23 (df = 252)	11.98 (df = 251)
F Statistic		156.10*** (df = 1; 251)
Note:		*p<0.1; **p<0.05; ***p<0.01

Baseline (OLS)



- As education goes up, income follows.
- Basic scatterplot and baseline model shows a strong, positive correlation
  - This model assumes <u>every county is independent</u>*
  - It assumes that what happens in Travis County has zero effect on Williamson County.
  - We know that's not how the real-world works.

- In a standard Ordinary Least Squares (OLS) regression, one of the core "Gauss-Markov" assumptions is that the observations (specifically the error terms) are independent of one another.
- When this assumption is met, the covariance between any two different error terms is zero: $Cov(\epsilon_i - \epsilon_j) = 0$ for $i \neq j$
- **Why Independence Matters?**
    - **Standard Errors**: If observations are correlated, OLS will typically underestimate the standard errors.
        - This leads to artificially low p-values and "false positives" (Type I errors)
    - **Reliability**: You might think a relationship is statistically significant when, in reality, you just have a bunch of repeating patterns in your data.
- Common Ways Independence is Violated
    - Autocorrelation (Time Series)
    - Clustering (Grouped Data)
    - Spatial Correlation



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$25
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40
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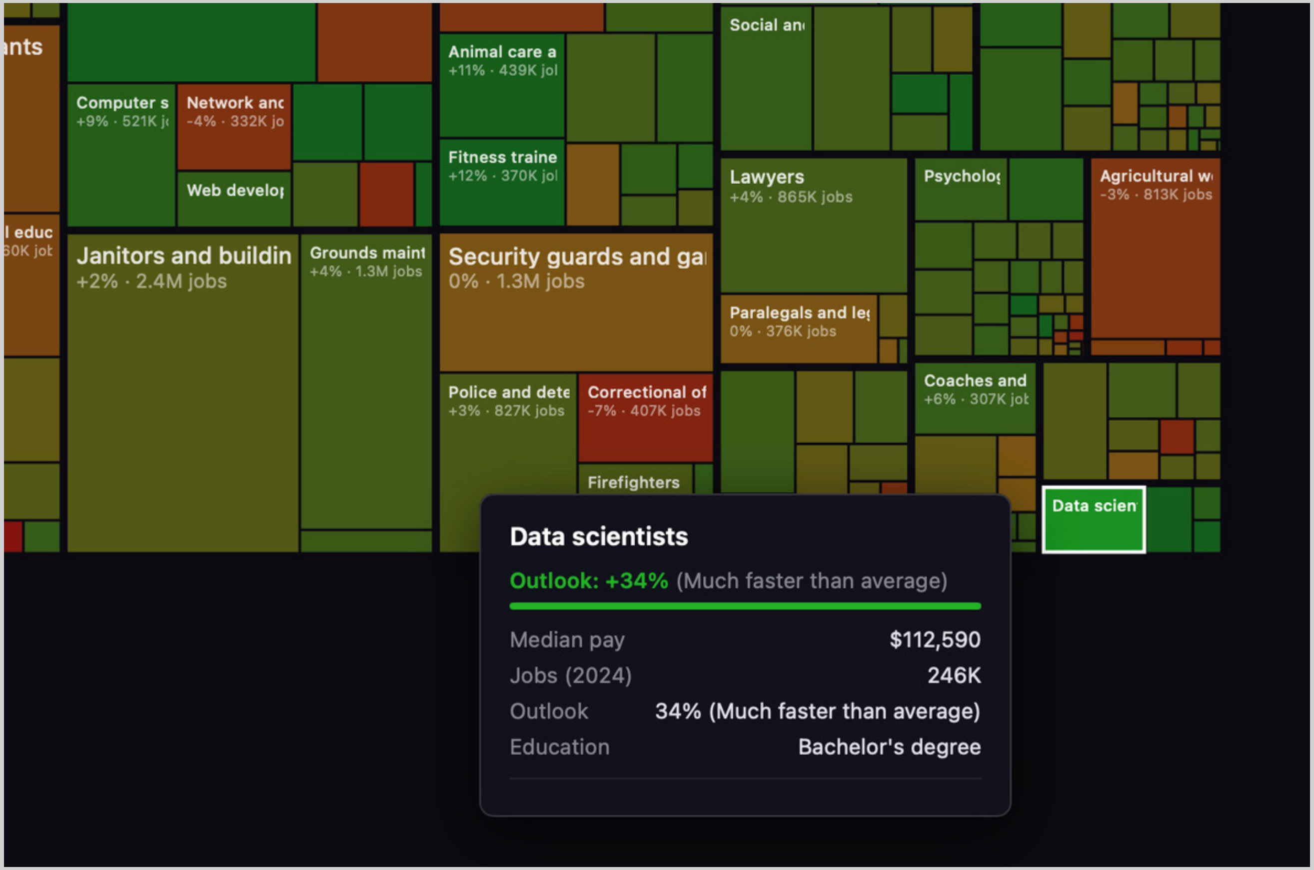
Percent With Bachelor Degree or Higher Education
Source: ACS 5-Year Data (2008-2022)
Brazos

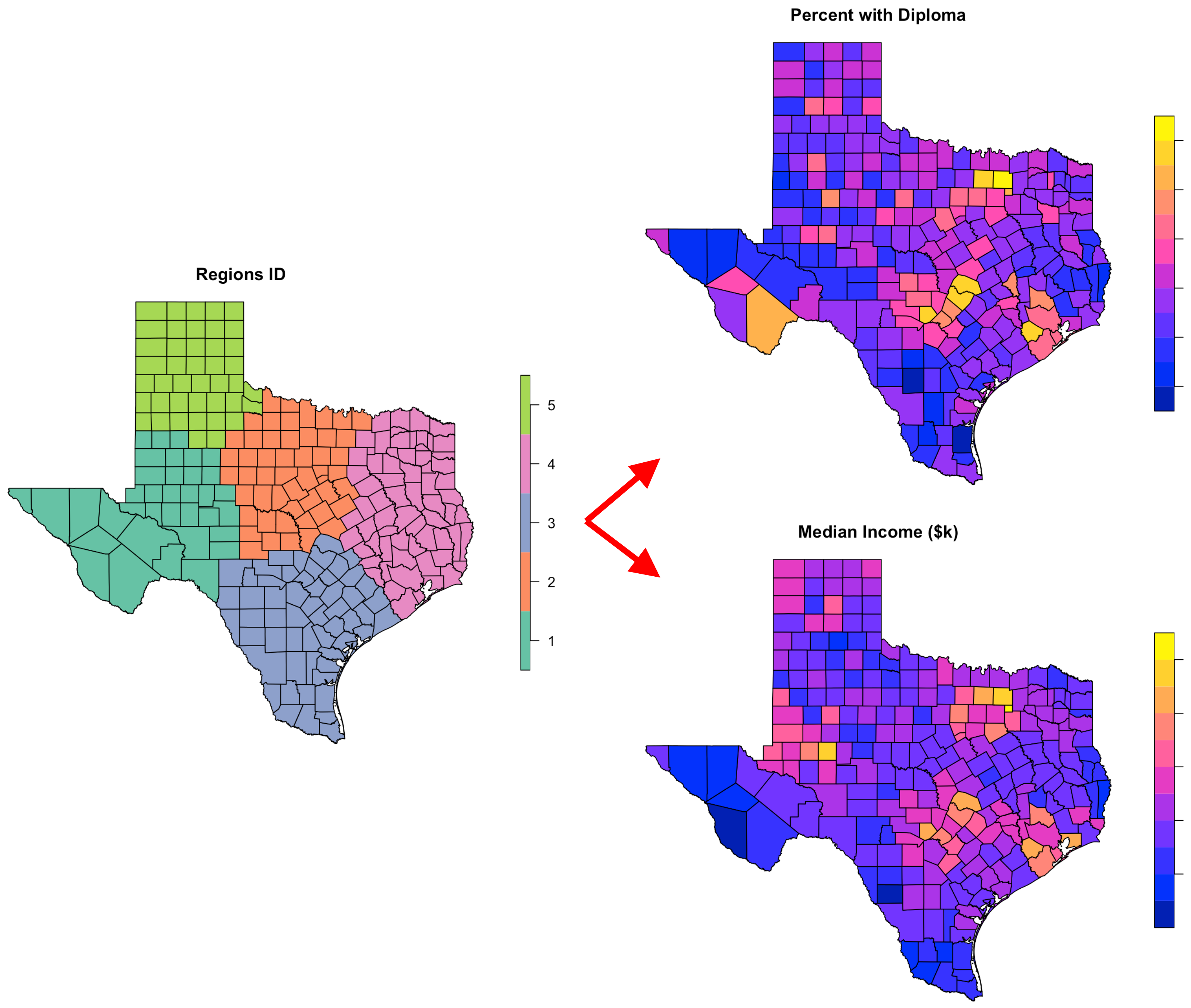
PS. Just sharing some labor statistics data:
- https://www.bls.gov/ooh/fastest-growing.htm
- https://www.bls.gov/ooh/math/data-scientists.htm
- https://www.bls.gov/ooh/architecture-and-engineering/industrial-engineers.htm

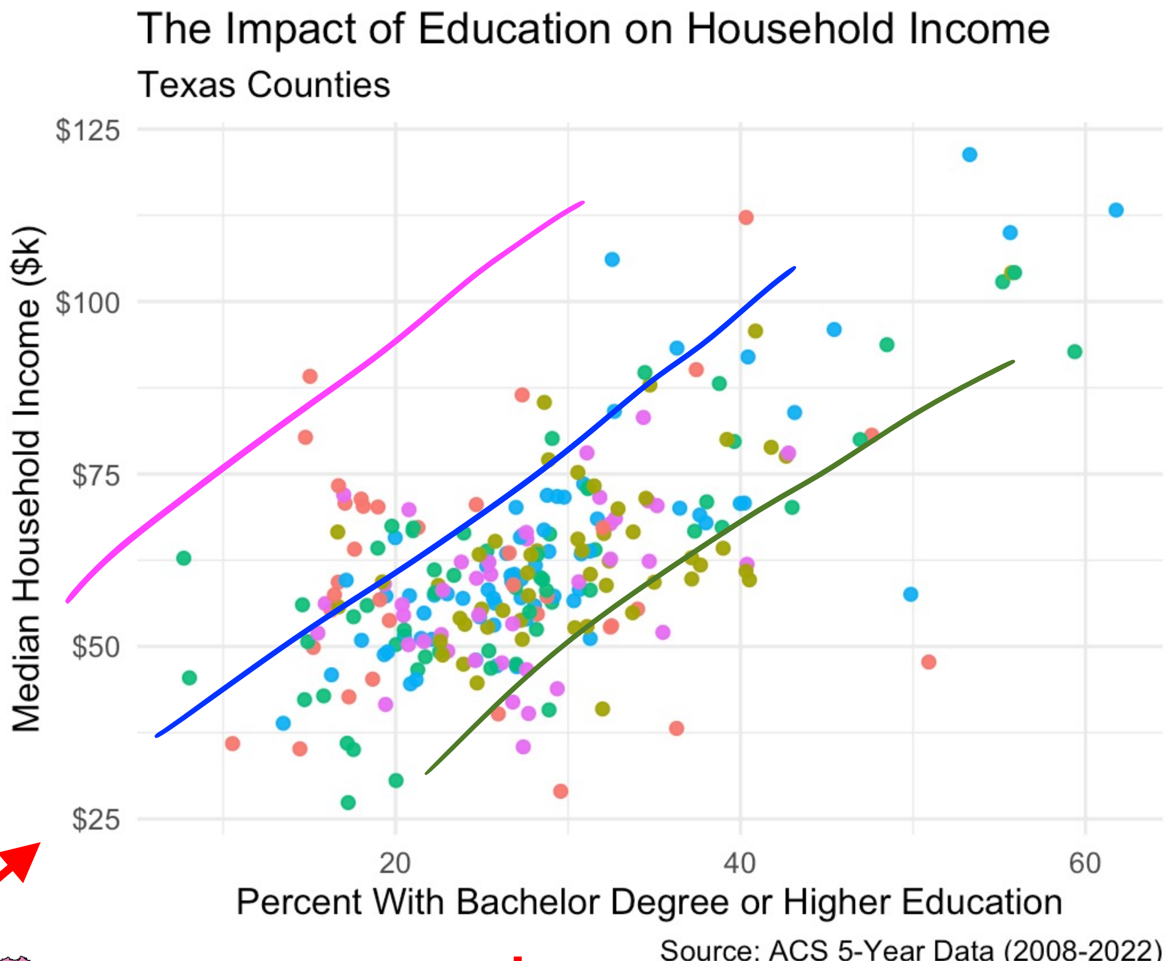
PS. This job market visualization (https://karpathy.ai/jobs/) is also interesting … helps to see the size of each area in the overall working force.



Question 2

- Geography could also be an influencer too …


Regions ID
5
4
3
2
1
Percent with Diploma
60
50
40
30
20
10
Median Income ($k)
120
100
80
60
40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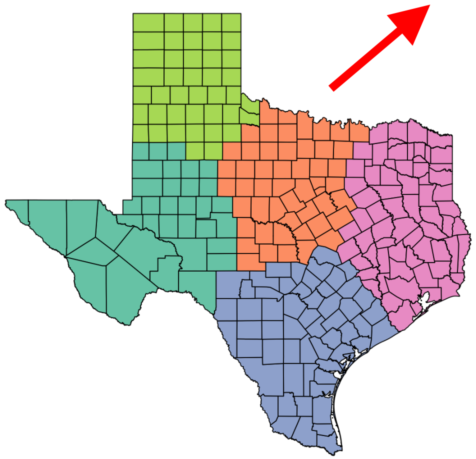
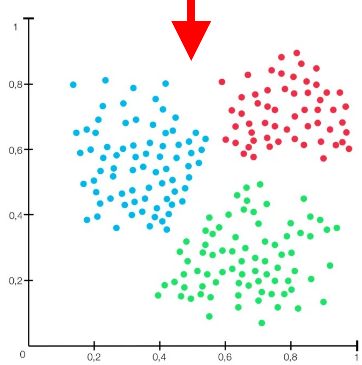



1
0,8
0,6
0,4
0,2
0
0,2
0,4
0,6
0,8
1

Multilevel Model

- Texas is a patchwork of different economies (Valley, Permian Basin, High Plains, etc.)
- A degree in a rural region might yield a different income than a degree in a tech hub.
- We use a **Random Intercept Model** to account for regional nesting:

$$Y_i = \beta_0 + \beta_1 X_{ij} + u_j + \epsilon_{ij}$$

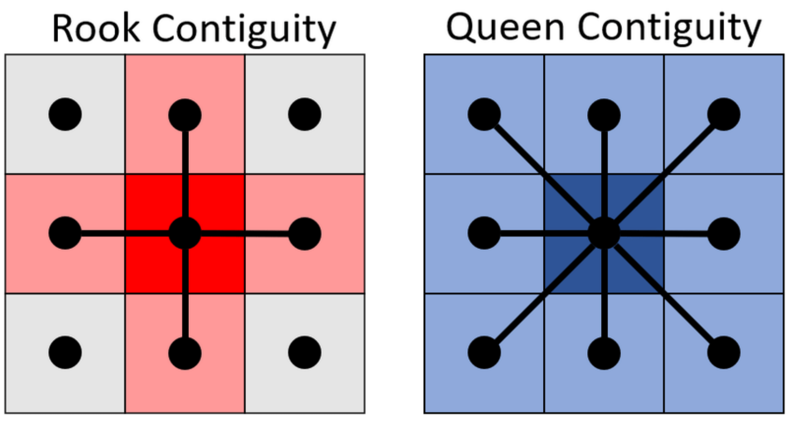
- We use K-means clustering to define 5 geographic regions (u_j)
- Goal: To see if the "Education Effect" stays the same once we acknowledge that different parts of Texas have different economic "baselines"

Multilevel Model

	Dependent variable:		
	med_income_k		
	OLS		linear mixed-effects
	(1)	(2)	(3)
pct_with_diploma		1.03***	1.03***
Constant	62.04***	33.13***	33.10***
Observations	253	253	253
R2	0.00	0.38	
Adjusted R2	0.00	0.38	
Log Likelihood			-987.20
Akaike Inf. Crit.			1,982.00
Bayesian Inf. Crit.			1,997.00
Residual Std. Error	15.23 (df = 252)	11.98 (df = 251)	11.97
Note:			*p<0.1; **p<0.05; ***p<0.01

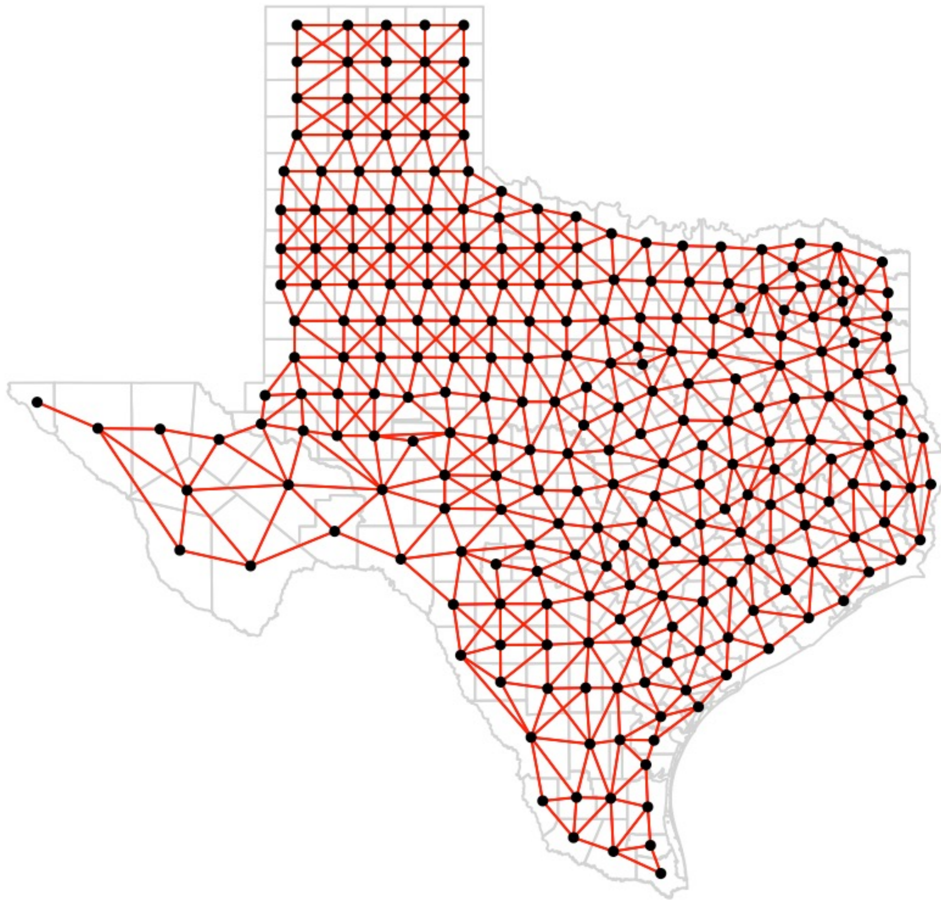
How do we define "nearby"?

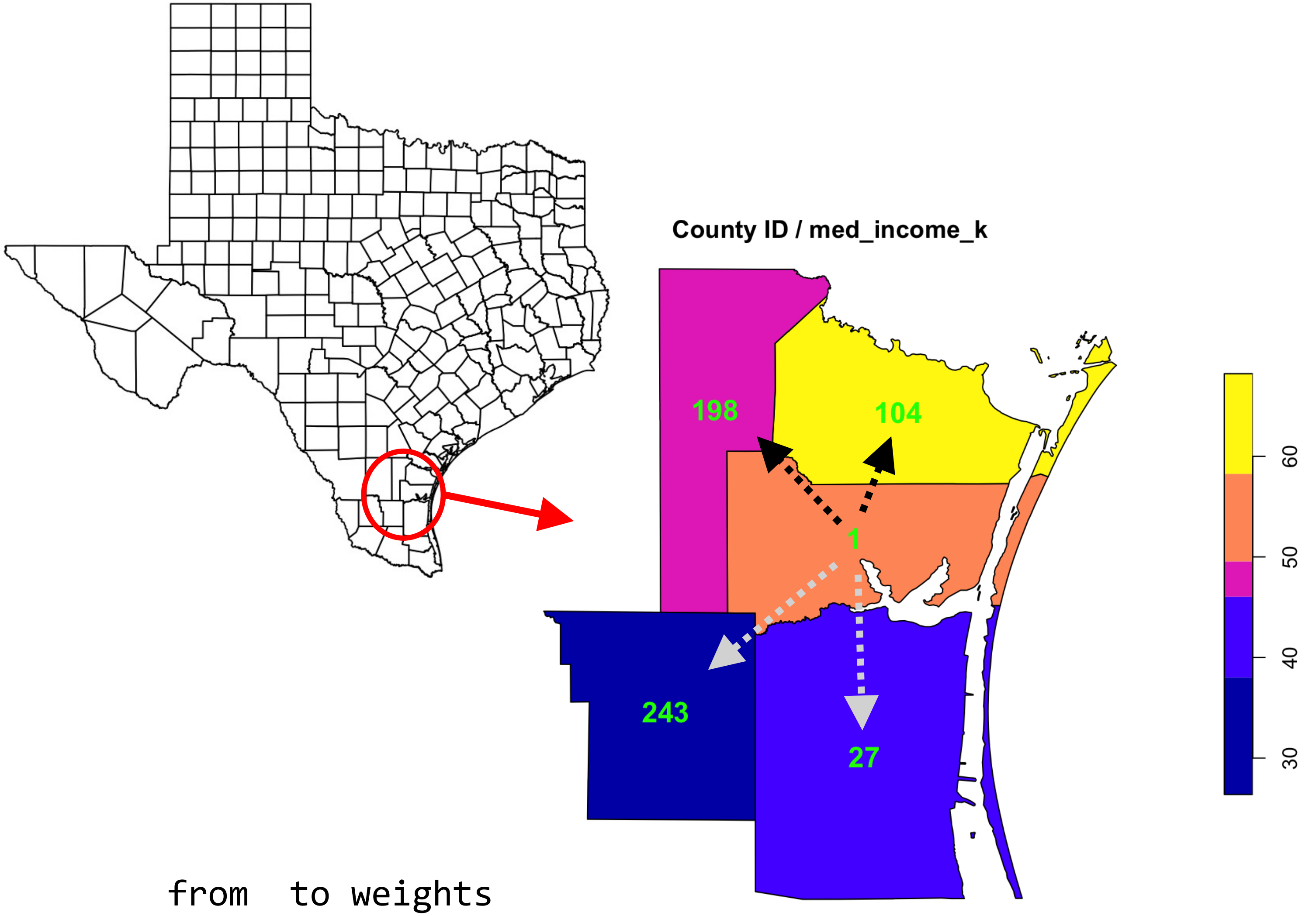
- In this analysis, let us use Queen Contiguity:
  - If two counties share even a single corner, they are neighbors (like a Queen on a chessboard)



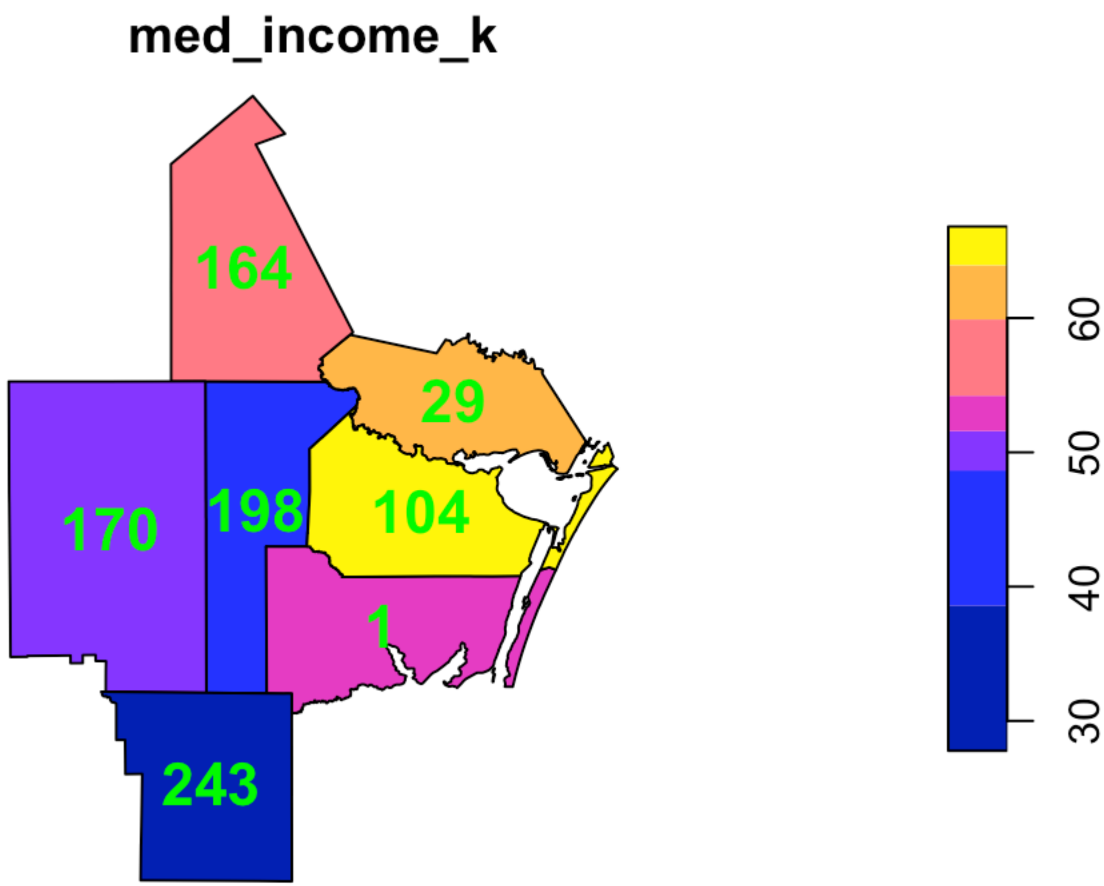
Spatial Weights Matrix (W)

- If county A and B are neighbors, $W_{ab} = 1$; otherwise, 0
  - Then **row-standardize** the matrix so the influence of all neighbors sums to 1
  - This allows us to calculate a "Spatial Lag" essentially the weighted average of everything happening next by





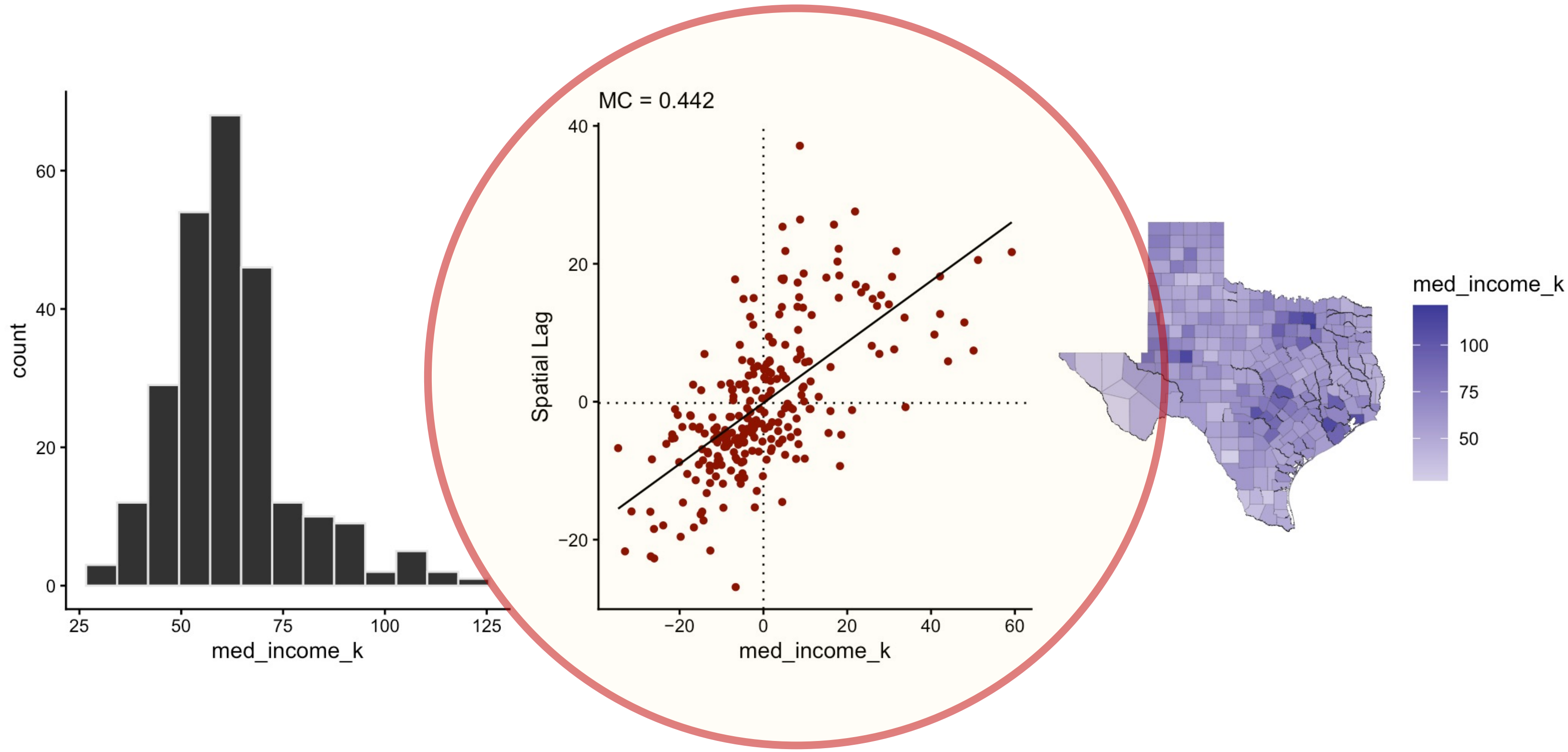
	from	to	weights
1	1	27	0.25
2	1	104	0.25
3	1	198	0.25
4	1	243	0.25




```
  from  to weights
1  198   1  0.1667
2  198  29  0.1667
3  198 104  0.1667
4  198 164  0.1667
5  198 170  0.1667
6  198 243  0.1667
```

Spatial Lag

- The spatial lag (y-axis) represents the average value of the variable in neighboring locations for each observation.
  - The scatterplot axis are centered around 0.
    - **Mean-Centering**: The function calculates the average for all counties.
    - **Subtraction**: then subtracts that average from every county value $(x_i - \bar{x})$.
  - **Spatial Lag**: is the weighted average of the centered values of a county's neighbors.



Local Indicators of Spatial Association (LISAs)

Local Geary: absolute difference between a county and its neighbors (dissimilarity).

Usually:

- **small** values of *Ci* indicate **positive spatial autocorrelation** (neighbors are similar)

- **large** values indicate **negative spatial autocorrelation** (neighbors are different)

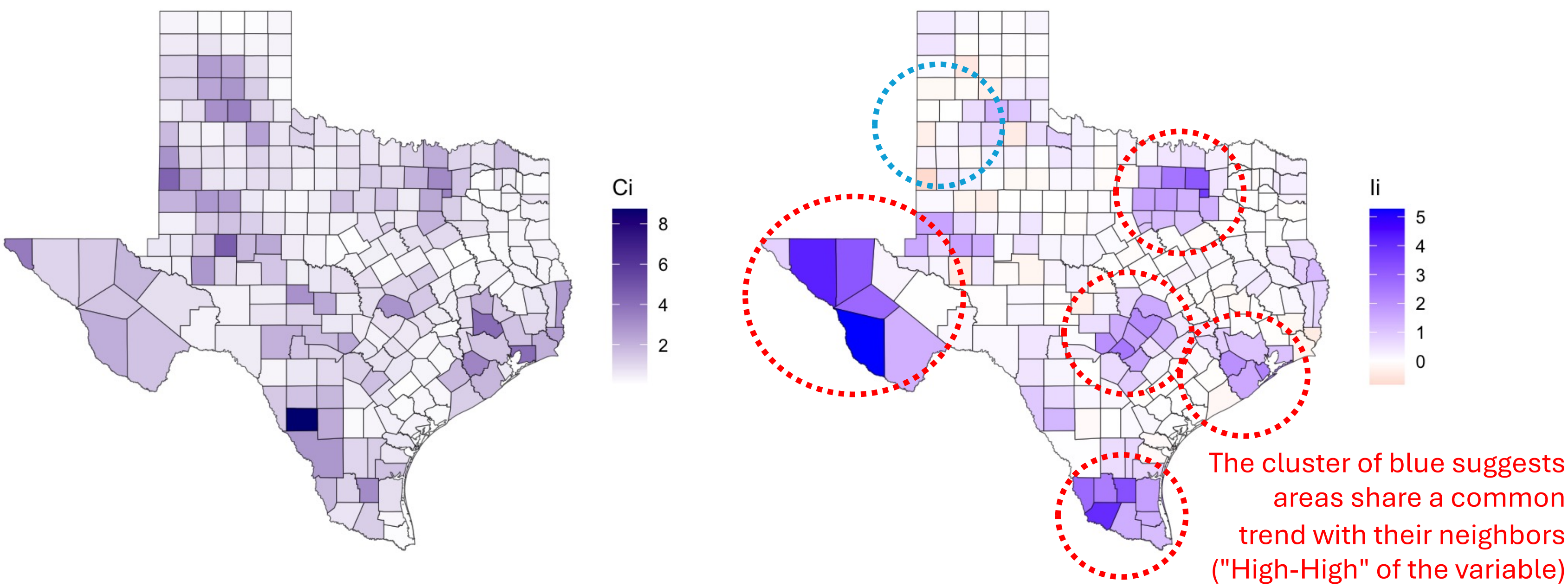
The dark blue areas represent counties that have very different values compared to their immediate neighbors. These are your spatial outliers:

Local Moran's I: cross-product of deviations from the mean. Measures whether a county and its neighbors are both above or both below the average.

- **Positive values** (**Blue**): Indicate Spatial Clusters. This means the county is surrounded by neighbors with similar values (High-High or Low-Low).

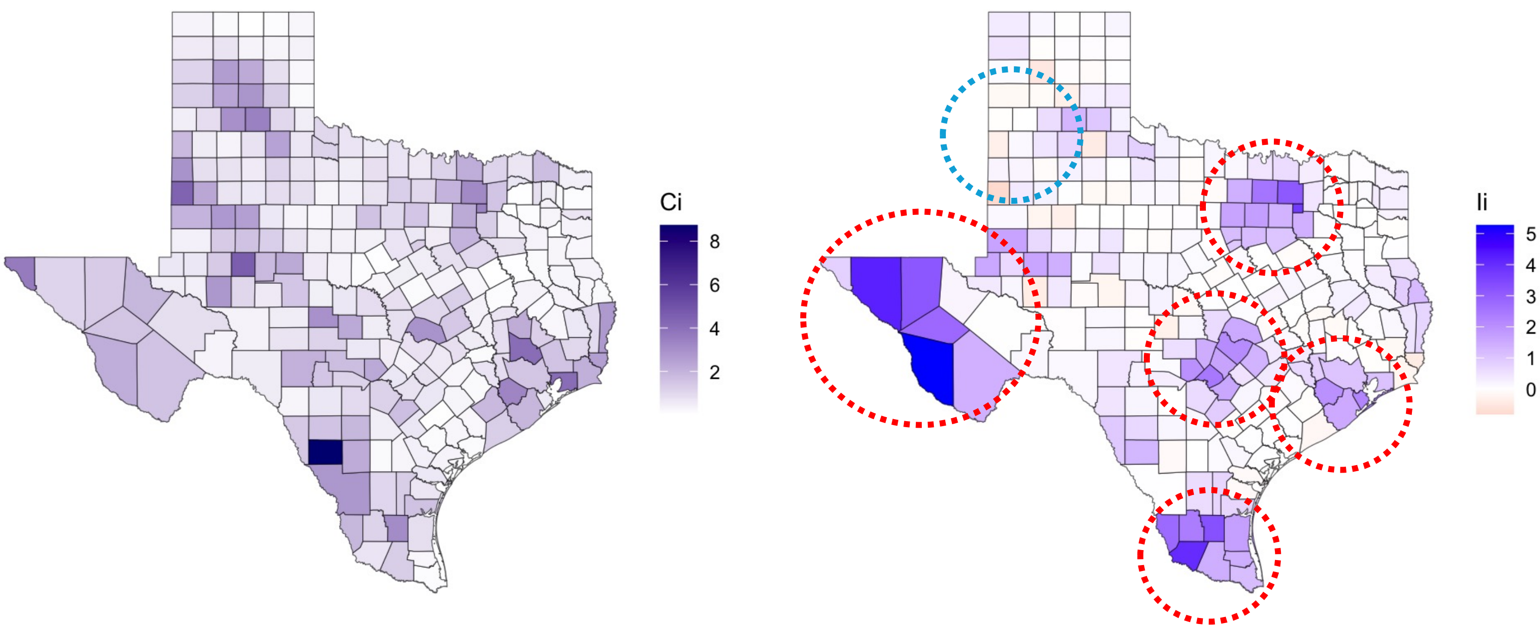
- **Near Zero** (**White**): Indicates a random spatial distribution.

- **Negative values** (**Red/Pink**): Indicate Spatial Outliers. A "High" value surrounded by "Low" values, or vice versa.



Local Indicators of Spatial Association (LISAs)

- **Indicates Clustering**: specific spots surrounded by similar values
  1. **High-High (Hot Spots)** A wealthy surrounded by wealthy
  2. **Low-Low (Cold Spots)** A poor surrounded by poor
  3. **High-Low (Spatial Outlier)**
  4. **Low-High (Spatial Outlier)**



Spatial Lag Model (SAR)

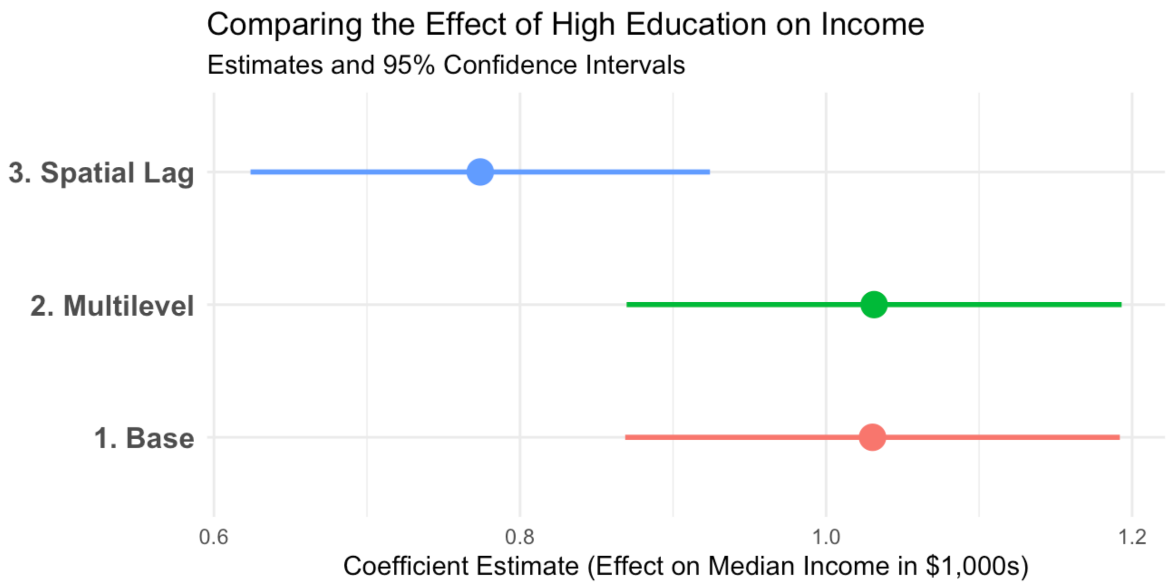
- We are no longer just looking at X (Education)
  we are looking at the *Spatial Lag* of Y:

$$Y = \rho WY + X\beta + \epsilon$$

- ρ (Rho): The autoregressive coefficient.
  - It tells us how much the income of neighboring counties "drags" your county's income up or down.
- WY: The "spatial lag": Represents the average income of the neighbors
- If ρ is significant, it means income is "contagious"

Multilevel Model

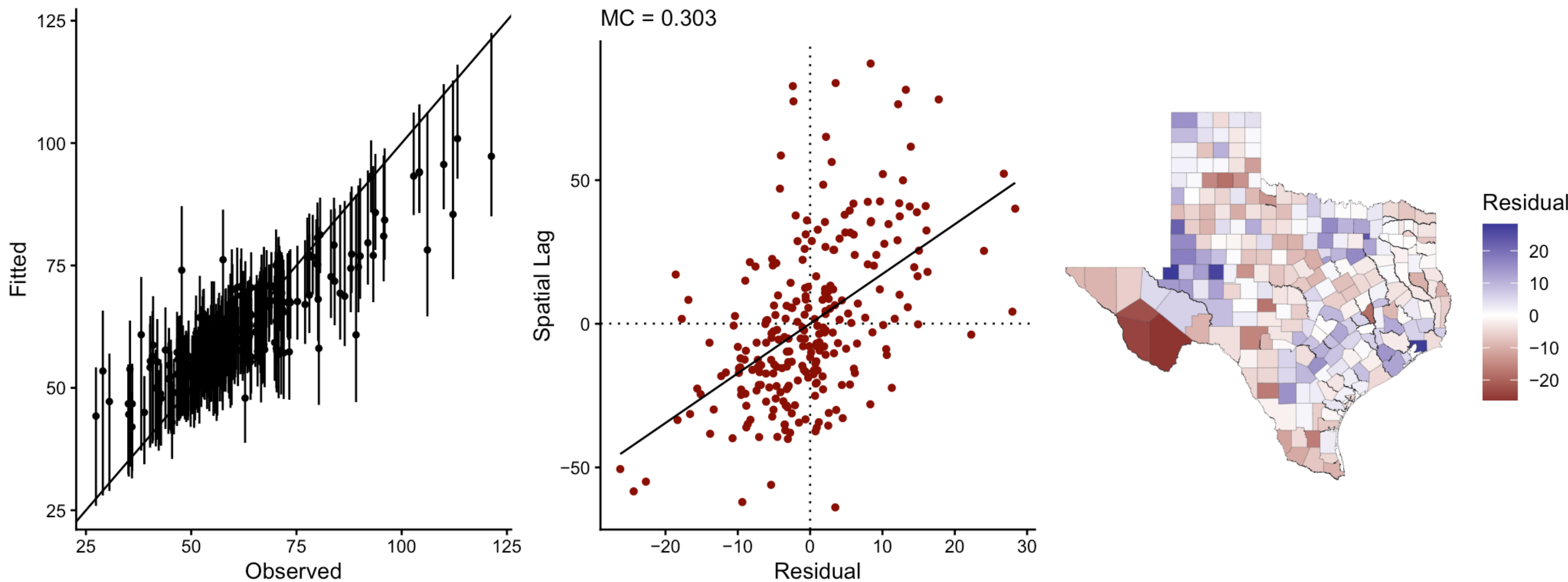
	Dependent variable:			
	med_income_k			
	OLS		linear mixed-effects	spatial autoregressive
	(1)	(2)	(3)	(4)
pct_with_diploma		1.03***	1.03***	0.77***
Constant	62.04***	33.13***	33.10***	7.58**
Observations	253	253	253	253
R2	0.00	0.38		
Adjusted R2	0.00	0.38		
Log Likelihood			-987.20	-953.20
Akaike Inf. Crit.			1,982.00	1,914.00
Residual Std. Error.	15.23 (df = 252)	11.98 (df = 251)	11.97	10.22
Note:				*p<0.1; **p<0.05; ***p<0.01



- In the simple OLS model:
  - "Education" was "stealing" the "credit" that actually belonged to the neighbors.
  - For instance: A county might be wealthy partly because of its degrees, but also because it's a bedroom community for a neighboring wealthy city.
- The Spatial Lag model separates these two:
  - The coefficient for Education gets smaller.
  - Tells us the direct effect of the degree versus the indirect effect of the location.

- Conclusions:
  - Education remains a primary driver of income across all models.
  - Being educated in a "wealthy cluster" is different than in a "spatial void".
  - Implication: To boost Texas economy, we can't just fund schools in isolation but foster regional ecosystems bc prosperity clearly spills over boundary lines.

0.0 = Perfectly random (Goal)
0.303 => **Clustered**



One more thing ("Model Stress Test")

- Where is the spatial model succeeding?
  Where is the "Geography of Texas" still outsmarting your math?
- Notice that as income gets higher (the right side of the graph),
  the error bars get longer and the dots drift further from the line.
- The model is good at predicting income for "average" counties,
  struggles with the extremely wealthy or extremely poor outliers.
  - 0.303 tells that the model errors are "contagious."
    - If the model under-predicts income in one county, it is very likely to under-predict it in the neighboring county too.
    - We have reduced the spatial bias, but we haven't eliminated it!

Learn More
Vignettes

- **Exploratory spatial data analysis**
  https://cran.r-project.org/web/packages/geostan/vignettes/measuring-sa.html
- **Spatial analysis** (with geostan)
  https://cran.r-project.org/web/packages/geostan/vignettes/spatial-analysis.html
- **Spatial measurement error models**
  https://cran.r-project.org/web/packages/geostan/vignettes/spatial-me-models.html
- **Custom spatial models** (with RStan and geostan)
  https://cran.r-project.org/web/packages/geostan/vignettes/custom-spatial-models.html
- **Raster regression**
  https://cran.r-project.org/web/packages/geostan/vignettes/raster-regression.html

References

- Spatial Data Science
  https://r-spatial.org/book/
  - Spatial Regression (Markov random field and multilevel models)
    https://r-spatial.org/book/16-SpatialRegression.html
  - Spatial Econometrics Models
    https://r-spatial.org/book/17-Econometrics.html